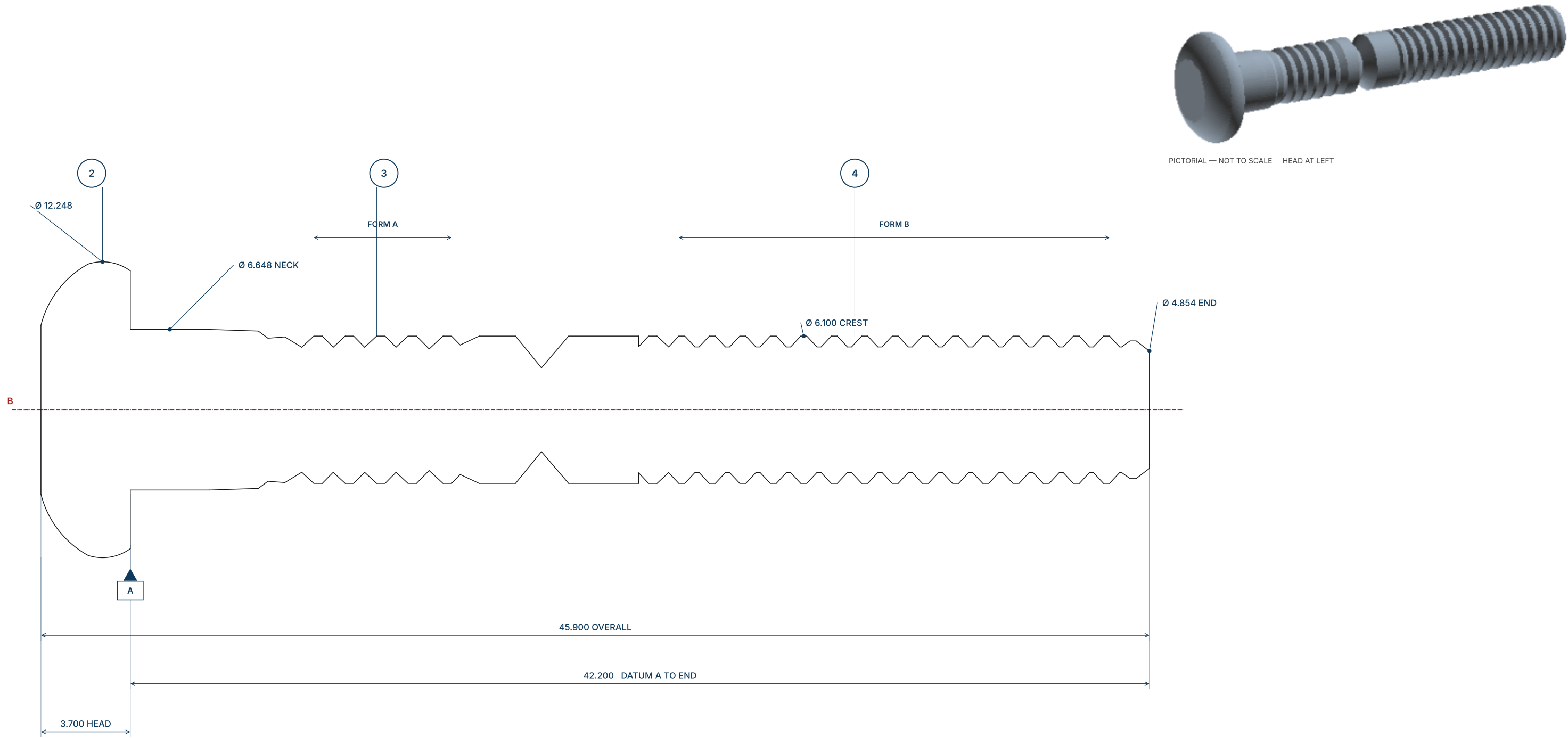




SCALE 6 : 1 AXIS HORIZONTAL HEAD AT LEFT FULL PROFILE

Solid of revolution about datum B. Revolve the +R profile.

Sizes are CAD nominals from 1 mid L.STEP (SolidWorks 2023, 2024-10-24).

No tolerance, material, or finish is stored in the model.

Graphic sizes are rounded to 0.001 mm. Sheet 6 lists every segment to 0.00001 mm.

Positive Z runs from datum A toward the shank end.

Volume 1493.044 mm³ area 1247.636 mm² centre of mass Z 14.862 on the axis.

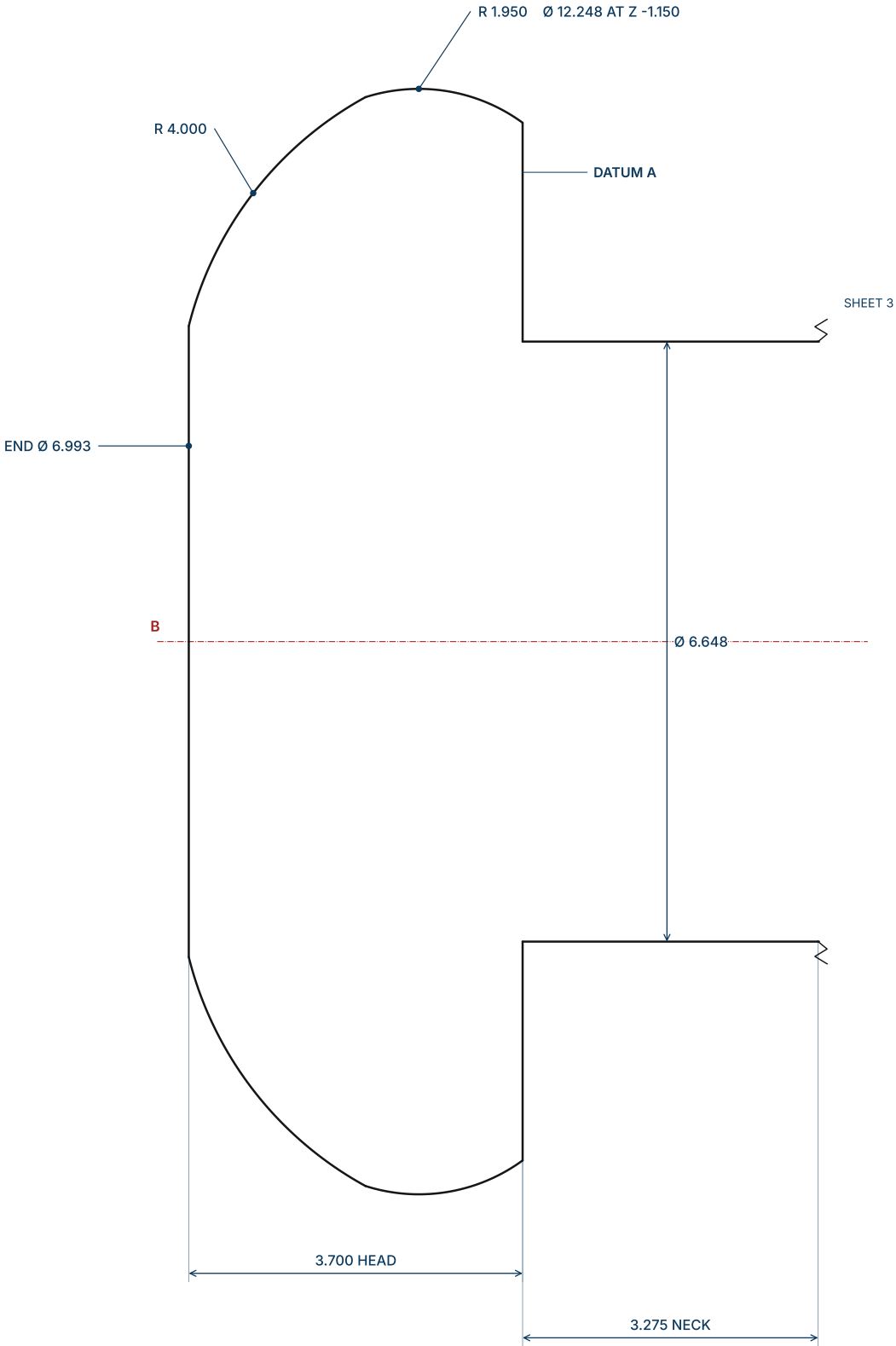
Balloons 2, 3 and 4 mark the detail sheets.

Form A: 5 crests, pitch 1.300 then 1.450. Form B: 15 crests, pitch 1.26250 then 1.25000.



PART	1 MID L	SHEET	1 / 6
FILE	1 mid L.STEP	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0	DATUM B AXIS		

DETAIL 2 SCALE 14 : 1 FULL PROFILE



HEAD STATIONS FROM DATUM A

POINT	Z	RADIUS	Ø	NOTE
HEAD FACE	-3.70000	0.00000	0.000	ON THE AXIS
HEAD RIM	-3.70000	3.49666	6.99332	R4 MEETS THE FACE
ARC JOIN	-1.74206	6.03219	12.06438	NOSE TO HEAD ARC
CROWN	-1.15000	6.12424	12.24848	LARGEST DIAMETER
DATUM A OUT	0.00000	5.74904	11.49809	SHOULDER OUTER
DATUM A IN	0.00000	3.32424	6.64848	SHOULDER INNER
NECK END	3.27487	3.32424	6.64848	LEAD-IN STARTS

AXIAL-SECTION ARCS — BOTH ARE TORUS SECTIONS

ARC	R	CENTER R	CENTER Z	ARC LENGTH
NOSE	4.00000	2.52424	0.18000	3.29596
HEAD	1.95000	4.17424	-1.15000	1.83150

Nose-arc center is inside the metal, 0.180 mm past datum A.

The head arc carries the largest diameter.

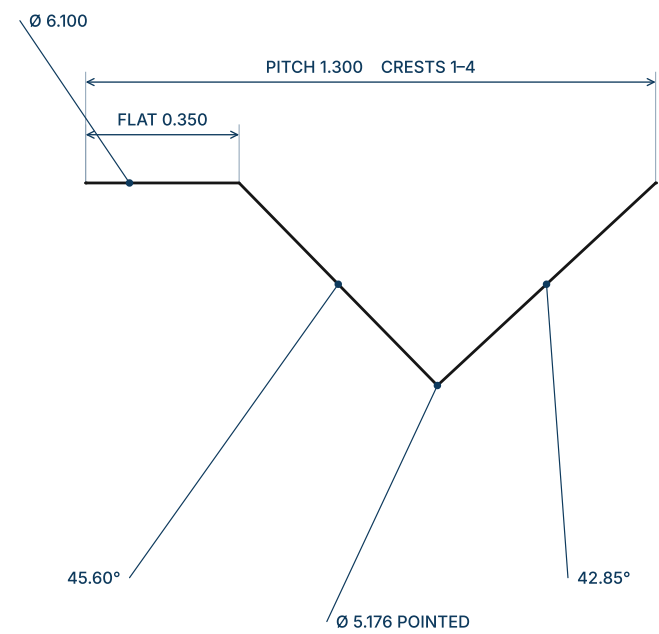
Datum A runs from the neck out to the head arc.

Lead-in taper, 1.87°, continues on sheet 3.

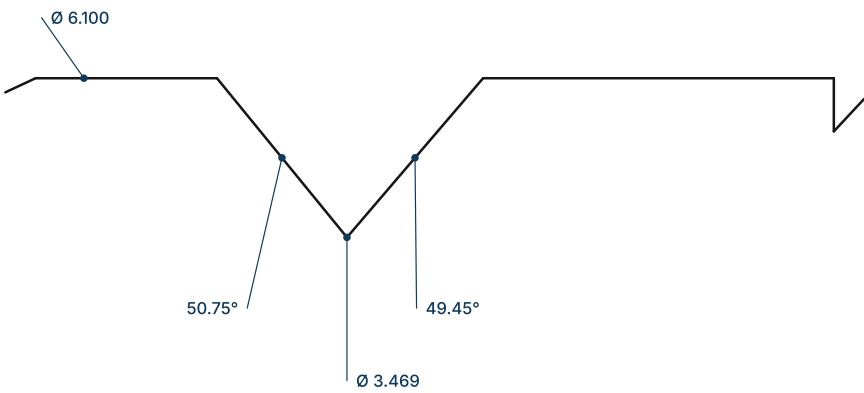


PART	1 MID L	SHEET	2 / 6
FILE	1 mid L.STEP	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0		DATUM B AXIS	

DETAIL 3 TOOTH AND GROOVE ARE ENLARGED ANGLES FROM THE AXIS



FORM A TOOTH SCALE 58 : 1 MATERIAL BELOW THE LINE CRESTS 1-3 SHARE THIS ROOT



LAND, GROOVE, BARREL, STEP SCALE 16 : 1 HALF PROFILE

LEAD-IN TAPERS, NECK END TO FIRST CREST

ANGLE	Z0	Z1	Ø START	Ø END
1.87°	3.27487	5.30000	6.64848	6.51625
36.60°	5.30000	5.70000	6.51625	5.92211
4.35°	5.70000	6.40000	5.92211	6.02861
31.45°	6.40000	7.10000	6.02861	5.17237
42.85°	7.10000	7.60000	5.17237	6.10000

FORM A CRESTS Ø 6.100 FLAT 0.350

Crests 1-4: fall 45.60°, pointed root Ø 5.176, rise 42.85°.

Crest 4 to 5 drops to Ø 5.031, so that pitch opens to 1.450.

#	Z0	Z1	FLAT	PITCH
1	7.60000	7.95000	0.35000	—
2	8.90000	9.25000	0.35000	1.30000
3	10.20000	10.55000	0.35000	1.30000
4	11.50000	11.85000	0.35000	1.30000
5	12.95000	13.30000	0.35000	1.45000

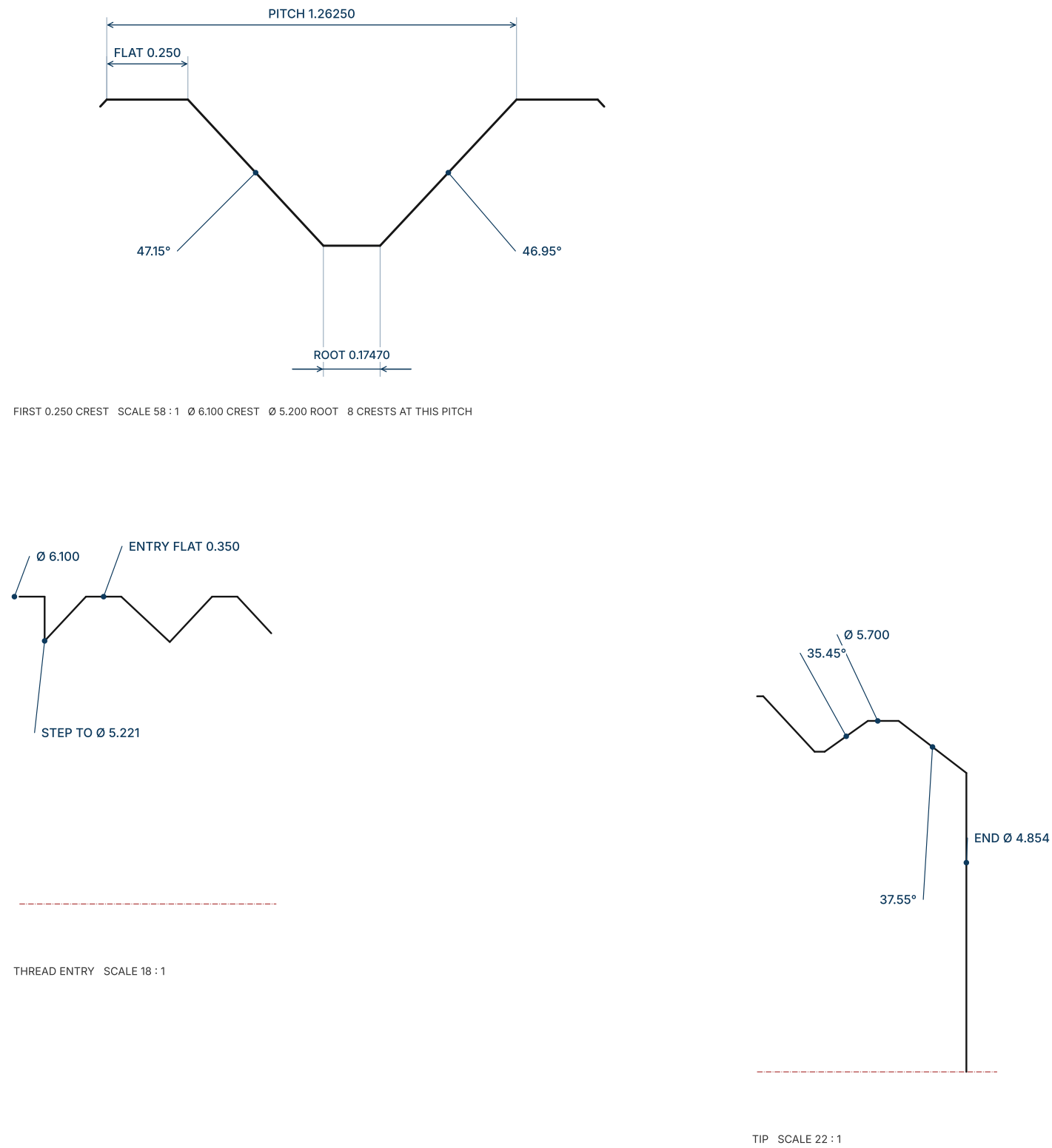
AFTER CREST 5

FEATURE	Z0	Z1	NOTE
25.05° FLANK	13.66108	14.45000	R 2.68128 TO 3.05000
LAND	14.45000	15.95000	L 1.500 Ø 6.100
GROOVE ROOT	17.02464	17.02464	R 1.73470 Ø 3.46940 POINTED
BARREL	18.15000	21.05000	L 2.900 Ø 6.100
STEP FACE	21.05000	21.05000	Ø 6.100 TO Ø 5.221



PART	1 MID L	SHEET	3 / 6
FILE	1 mid L.STEP	UNITS	MILLIMETRES
DATUM A	SHOULDER Z = 0	DATUM B	AXIS

DETAIL 4 REGULAR CREST ENLARGED ENTRY AND TIP AT TRUE RADIAL SCALE

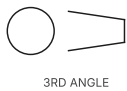


FORM B — 15 CRESTS, FLAT 0.250, Ø 6.100

#	Z0	Z1	PITCH	#	Z0	Z1	PITCH
1	22.71250	22.96250	—	9	32.80000	33.05000	1.25000
2	23.97500	24.22500	1.26250	10	34.05000	34.30000	1.25000
3	25.23750	25.48750	1.26250	11	35.30000	35.55000	1.25000
4	26.50000	26.75000	1.26250	12	36.55000	36.80000	1.25000
5	27.76250	28.01250	1.26250	13	37.80000	38.05000	1.25000
6	29.02500	29.27500	1.26250	14	39.05000	39.30000	1.25000
7	30.28750	30.53750	1.26250	15	40.30000	40.55000	1.25000
8	31.55000	31.80000	1.26250				

DEFINITION

ITEM	VALUE	NOTE
CREST Ø / FLAT	6.100 / 0.250	15 CRESTS AFTER THE ENTRY
ENTRY CREST	Z 21.46041	FLAT 0.35000 NOT IN THE 15
ROOT Ø	5.200	AXIAL FLAT, SEE GROUPS
FALL / RISE	47.15° / 46.95°	FROM THE AXIS
PITCH CHANGE	AFTER CREST 8	1.26250 THEN 1.25000
ROOT ×7	0.17470	Z 23.37993 ... 31.12963
ROOT ×7	0.16220	Z 32.21743 ... 39.87963
ROOT ×1	0.08143	Z 40.96743 ... 41.04887



PART	1 MID L	SHEET	4 / 6
FILE	1 mid L.STEP	UNITS	MILLIMETRES
DATUM A SHOULDERS Z = 0		DATUM B AXIS	

CONTROLLING SIZES: Z FROM DATUM A, POSITIVE TOWARD THE SHANK. ANGLES FROM THE AXIS. MODEL VALUES TO 0.00001 mm.

1 OVERALL

ITEM	MODEL	NOTE
OVERALL	45.90000	HEAD FACE TO END FACE
HEAD FACE Z	-3.70000	Ø 6.99332
DATUM A	0.00000	Ø 6.64848 TO 11.49809
LARGEST Ø	12.24848	AT Z -1.15000
END FACE Z	42.20000	Ø 4.85441
VOLUME	1493.04354	mm ³ SOLID, NO BORE
SURFACE	1247.63569	mm ²
CON Z	14.86214	ON THE AXIS

2 ARCS

ARC	R	CTR R	CTR Z	Z0	Z1
NOSE	4.00000	2.52424	0.18000	-3.70000	-1.74206
HEAD	1.95000	4.17424	-1.15000	-1.74206	0.00000

3 NECK

ITEM	Z0	Z1	SIZE
CYLINDER	0.00000	3.27487	Ø 6.64848 L 3.27487

4 AXIS IN THE STEP FILE

ITEM	X	Y	Z
DIRECTION	0.000538	0.999970	0.007686
POINT ON AXIS	0.09696	42.19848	0.35385

The point lies in the shank end plane. Z above is the STEP frame, not datum A.

5 EVERY CONTROLLING FEATURE

ITEM	Z0	Z1	SIZE
LEAD 1.87°	3.27487	5.30000	Ø 6.64848 → 6.51625
LEAD 36.60°	5.30000	5.70000	Ø 6.51625 → 5.92211
LEAD 4.35°	5.70000	6.40000	Ø 5.92211 → 6.02861
LEAD 31.45°	6.40000	7.10000	Ø 6.02861 → 5.17237
LEAD 42.85°	7.10000	7.60000	Ø 5.17237 → 6.10000
A CREST 1	7.60000	7.95000	Ø 6.10000 FLAT 0.35000
A CREST 2	8.90000	9.25000	Ø 6.10000 FLAT 0.35000
A CREST 3	10.20000	10.55000	Ø 6.10000 FLAT 0.35000
A CREST 4	11.50000	11.85000	Ø 6.10000 FLAT 0.35000
A CREST 5	12.95000	13.30000	Ø 6.10000 FLAT 0.35000
FLANK 25.05°	13.66108	14.45000	R 2.68128 → 3.05000
LAND	14.45000	15.95000	Ø 6.10000
GROOVE 50.75°	15.95000	17.02464	TO R 1.73470
GROOVE 49.45°	17.02464	18.15000	TO Ø 6.10000
BARREL	18.15000	21.05000	Ø 6.10000
STEP	21.05000	21.05000	Ø 6.10000 → 5.22131
B ENTRY	21.46041	21.81041	FLAT 0.35000
B CREST 1	22.71250	22.96250	PITCH GROUP 1.26250
B CREST 8	31.55000	31.80000	LAST OF PITCH 1.26250
B CREST 9	32.80000	33.05000	FIRST OF PITCH 1.25000
B CREST 15	40.30000	40.55000	LAST 0.250 FLAT
ROOT ×7	23.37993	31.12963	FLAT 0.17470 Ø 5.20000
ROOT ×7	32.21743	39.87963	FLAT 0.16220 Ø 5.20000
ROOT ×1	40.96743	41.04887	FLAT 0.08143 Ø 5.20000
TIP 35.45°	41.04887	41.40000	Ø 5.20000 → 5.70000
TIP LAND	41.40000	41.65000	Ø 5.70000
TIP 37.55°	41.65000	42.20000	TO Ø 4.85441
END FACE	42.20000	42.20000	Ø 4.85441 TO THE AXIS



SHEET 6 LISTS ALL 97 PROFILE SEGMENTS, INCLUDING EVERY FLANK BETWEEN THE CRESTS ABOVE.

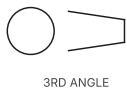
PART	1 MID L	SHEET	5 / 6
FILE	1 mid L.STEP	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0		DATUM B AXIS	

ORDERED +R PROFILE, HEAD FACE TO SHANK END. REVOLVE ABOUT DATUM B. MILLIMETRES TO 0.00001. ANGLE IS FROM THE AXIS.

#	KIND	Z1	R1	Z2	R2	LENGTH	ANGLE	ARC
001	FACE	-3.70000	0.00000	-3.70000	3.49666	3.49666	90.00	
002	ARC	-3.70000	3.49666	-1.74206	6.03219	3.29596		R4.00000
003	ARC	-1.74206	6.03219	0.00000	5.74904	1.83150		R1.95000
004	FACE	0.00000	5.74904	0.00000	3.32424	2.42480	90.00	
005	CYL	0.00000	3.32424	3.27487	3.32424	3.27487	0.00	
006	TAPER	3.27487	3.32424	5.30000	3.25812	2.02621	1.87	
007	TAPER	5.30000	3.25812	5.70000	2.96106	0.49825	36.60	
008	TAPER	5.70000	2.96106	6.40000	3.01430	0.70202	4.35	
009	TAPER	6.40000	3.01430	7.10000	2.58618	0.82054	31.45	
010	TAPER	7.10000	2.58618	7.60000	3.05000	0.68200	42.85	
011	CYL	7.60000	3.05000	7.95000	3.05000	0.35000	0.00	
012	TAPER	7.95000	3.05000	8.40220	2.58023	0.64631	45.60	
013	TAPER	8.40220	2.58023	8.90000	3.05000	0.67900	42.85	
014	CYL	8.90000	3.05000	9.25000	3.05000	0.35000	0.00	
015	TAPER	9.25000	3.05000	9.70220	2.58023	0.64631	45.60	
016	TAPER	9.70220	2.58023	10.20000	3.05000	0.67900	42.85	
017	CYL	10.20000	3.05000	10.55000	3.05000	0.35000	0.00	
018	TAPER	10.55000	3.05000	11.00220	2.58023	0.64631	45.60	
019	TAPER	11.00220	2.58023	11.50000	3.05000	0.67900	42.85	
020	CYL	11.50000	3.05000	11.85000	3.05000	0.35000	0.00	
021	TAPER	11.85000	3.05000	12.37360	2.51531	0.74836	45.60	
022	TAPER	12.37360	2.51531	12.95000	3.05000	0.78621	42.85	
023	CYL	12.95000	3.05000	13.30000	3.05000	0.35000	0.00	
024	TAPER	13.30000	3.05000	13.66108	2.60128	0.51607	45.60	
025	TAPER	13.66108	2.60128	14.45000	3.05000	0.87004	25.05	
026	CYL	14.45000	3.05000	15.95000	3.05000	1.50000	0.00	
027	TAPER	15.95000	3.05000	17.02464	1.73470	1.69849	50.75	
028	TAPER	17.02464	1.73470	18.15000	3.05000	1.73102	49.45	
029	CYL	18.15000	3.05000	21.05000	3.05000	2.90000	0.00	
030	FACE	21.05000	3.05000	21.05000	2.61066	0.43934	90.00	
031	TAPER	21.05000	2.61066	21.46041	3.05000	0.60121	46.95	
032	CYL	21.46041	3.05000	21.81041	3.05000	0.35000	0.00	
033	TAPER	21.81041	3.05000	22.29213	2.60000	0.65921	43.05	
034	TAPER	22.29213	2.60000	22.71250	3.05000	0.61580	46.95	
035	CYL	22.71250	3.05000	22.96250	3.05000	0.25000	0.00	
036	TAPER	22.96250	3.05000	23.37993	2.60000	0.61380	47.15	
037	CYL	23.37993	2.60000	23.55463	2.60000	0.17470	0.00	
038	TAPER	23.55463	2.60000	23.97500	3.05000	0.61580	46.95	
039	CYL	23.97500	3.05000	24.22500	3.05000	0.25000	0.00	
040	TAPER	24.22500	3.05000	24.64243	2.60000	0.61380	47.15	
041	CYL	24.64243	2.60000	24.81713	2.60000	0.17470	0.00	
042	TAPER	24.81713	2.60000	25.23750	3.05000	0.61580	46.95	
043	CYL	25.23750	3.05000	25.48750	3.05000	0.25000	0.00	
044	TAPER	25.48750	3.05000	25.90493	2.60000	0.61380	47.15	
045	CYL	25.90493	2.60000	26.07963	2.60000	0.17470	0.00	
046	TAPER	26.07963	2.60000	26.50000	3.05000	0.61580	46.95	
047	CYL	26.50000	3.05000	26.75000	3.05000	0.25000	0.00	
048	TAPER	26.75000	3.05000	27.16743	2.60000	0.61380	47.15	
049	CYL	27.16743	2.60000	27.34213	2.60000	0.17470	0.00	

#	KIND	Z1	R1	Z2	R2	LENGTH	ANGLE	ARC
050	TAPER	27.34213	2.60000	27.76250	3.05000	0.61580	46.95	
051	CYL	27.76250	3.05000	28.01250	3.05000	0.25000	0.00	
052	TAPER	28.01250	3.05000	28.42993	2.60000	0.61380	47.15	
053	CYL	28.42993	2.60000	28.60463	2.60000	0.17470	0.00	
054	TAPER	28.60463	2.60000	29.02500	3.05000	0.61580	46.95	
055	CYL	29.02500	3.05000	29.27500	3.05000	0.25000	0.00	
056	TAPER	29.27500	3.05000	29.69243	2.60000	0.61380	47.15	
057	CYL	29.69243	2.60000	29.86713	2.60000	0.17470	0.00	
058	TAPER	29.86713	2.60000	30.28750	3.05000	0.61580	46.95	
059	CYL	30.28750	3.05000	30.53750	3.05000	0.25000	0.00	
060	TAPER	30.53750	3.05000	30.95493	2.60000	0.61380	47.15	
061	CYL	30.95493	2.60000	31.12963	2.60000	0.17470	0.00	
062	TAPER	31.12963	2.60000	31.55000	3.05000	0.61580	46.95	
063	CYL	31.55000	3.05000	31.80000	3.05000	0.25000	0.00	
064	TAPER	31.80000	3.05000	32.21743	2.60000	0.61380	47.15	
065	CYL	32.21743	2.60000	32.37963	2.60000	0.16220	0.00	
066	TAPER	32.37963	2.60000	32.80000	3.05000	0.61580	46.95	
067	CYL	32.80000	3.05000	33.05000	3.05000	0.25000	0.00	
068	TAPER	33.05000	3.05000	33.46743	2.60000	0.61380	47.15	
069	CYL	33.46743	2.60000	33.62963	2.60000	0.16220	0.00	
070	TAPER	33.62963	2.60000	34.05000	3.05000	0.61580	46.95	
071	CYL	34.05000	3.05000	34.30000	3.05000	0.25000	0.00	
072	TAPER	34.30000	3.05000	34.71743	2.60000	0.61380	47.15	
073	CYL	34.71743	2.60000	34.87963	2.60000	0.16220	0.00	
074	TAPER	34.87963	2.60000	35.30000	3.05000	0.61580	46.95	
075	CYL	35.30000	3.05000	35.55000	3.05000	0.25000	0.00	
076	TAPER	35.55000	3.05000	35.96743	2.60000	0.61380	47.15	
077	CYL	35.96743	2.60000	36.12963	2.60000	0.16220	0.00	
078	TAPER	36.12963	2.60000	36.55000	3.05000	0.61580	46.95	
079	CYL	36.55000	3.05000	36.80000	3.05000	0.25000	0.00	
080	TAPER	36.80000	3.05000	37.21743	2.60000	0.61380	47.15	
081	CYL	37.21743	2.60000	37.37963	2.60000	0.16220	0.00	
082	TAPER	37.37963	2.60000	37.80000	3.05000	0.61580	46.95	
083	CYL	37.80000	3.05000	38.05000	3.05000	0.25000	0.00	
084	TAPER	38.05000	3.05000	38.46743	2.60000	0.61380	47.15	
085	CYL	38.46743	2.60000	38.62963	2.60000	0.16220	0.00	
086	TAPER	38.62963	2.60000	39.05000	3.05000	0.61580	46.95	
087	CYL	39.05000	3.05000	39.30000	3.05000	0.25000	0.00	
088	TAPER	39.30000	3.05000	39.71743	2.60000	0.61380	47.15	
089	CYL	39.71743	2.60000	39.87963	2.60000	0.16220	0.00	
090	TAPER	39.87963	2.60000	40.30000	3.05000	0.61580	46.95	
091	CYL	40.30000	3.05000	40.55000	3.05000	0.25000	0.00	
092	TAPER	40.55000	3.05000	40.96743	2.60000	0.61380	47.15	
093	CYL	40.96743	2.60000	41.04807	2.60000	0.08143	0.00	
094	TAPER	41.04807	2.60000	41.40000	2.85000	0.43104	35.45	
095	CYL	41.40000	2.85000	41.65000	2.85000	0.25000	0.00	
096	TAPER	41.65000	2.85000	42.20000	2.42721	0.69372	37.55	
097	FACE	42.20000	2.42721	42.20000	0.00000	2.42721	90.00	

ARC 002 R 4.00000 CENTER R 2.52424 CENTER Z 0.18000 ARC 003 R 1.95000 CENTER R 4.17424 CENTER Z -1.15000



PART	1 MID L	SHEET	6 / 6
FILE	1 mid L.STEP	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0		DATUM B AXIS	